AI FOR ROBOTICS · FINAL PROJECT, TRACK 1

Decision Transformer and the stitching problem

Reproducing the paper, then improving it with expectile-based return relabeling

Based on Chen et al., Decision Transformer, NeurIPS 2021

01 · MOTIVATION

Real robots can’t afford endless trial and error

Slow

Every episode runs in real time and wears the hardware.

Risky

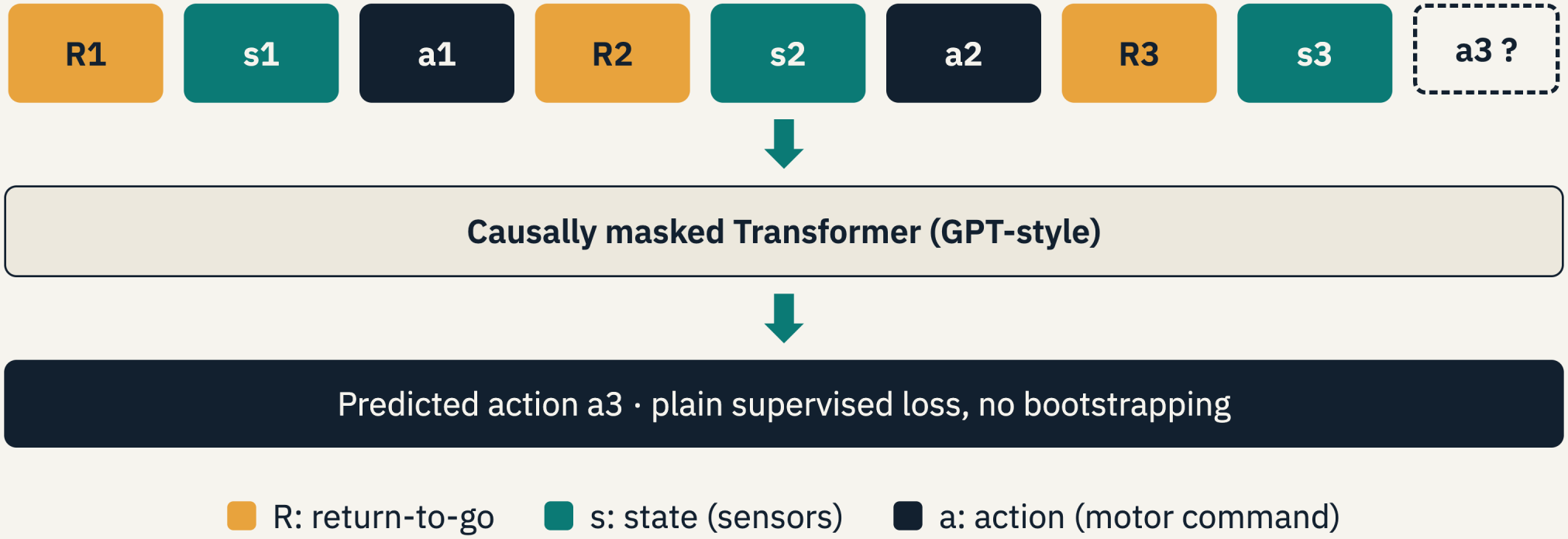
Random exploration can damage the robot and its surroundings.

Already logged

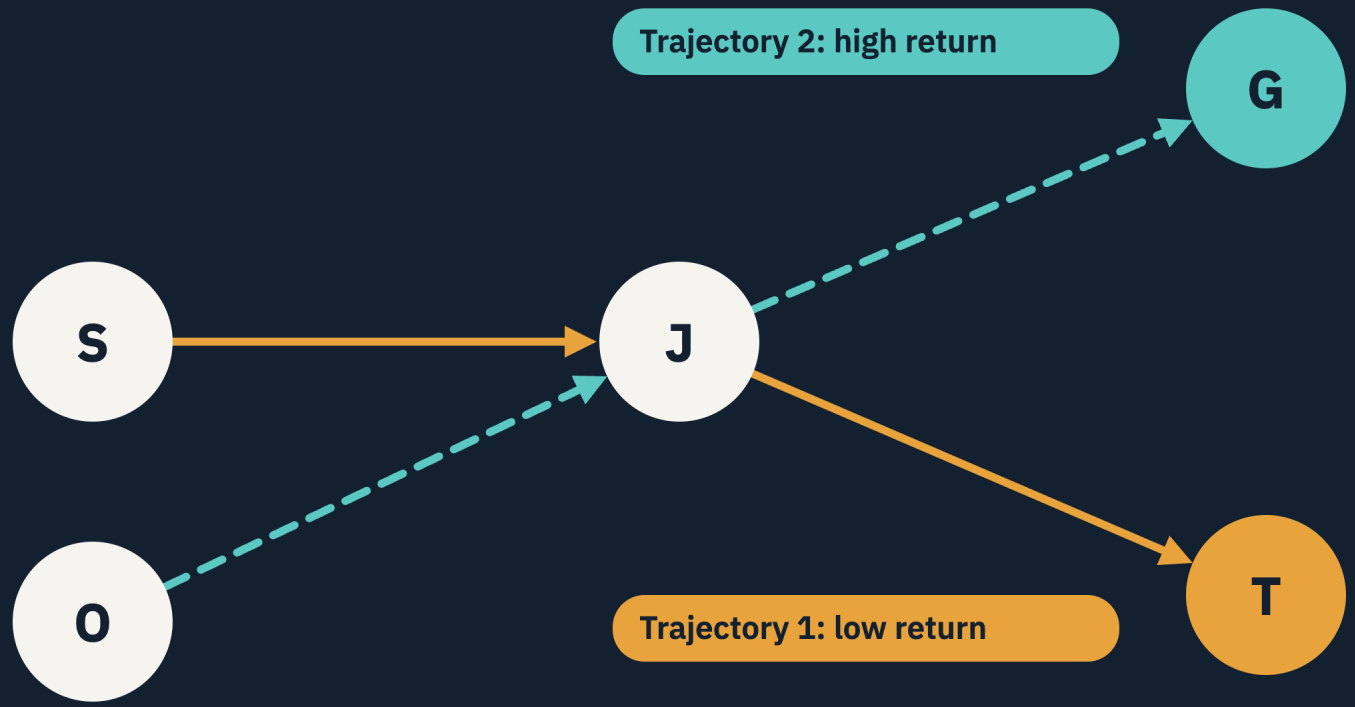
Labs hold logs, demonstrations and past controllers.

Offline RL: learn the best policy you can from a fixed dataset, with no new interaction.

Decision Transformer: RL as sequence modeling



The weakness: DT struggles to stitch



Why DT misses the best path

Each trajectory carries one label: its own total return.

The good first half of trajectory 1 inherits a low label, so DT struggles to tell it is worth keeping.

Best path: S → J → G, never seen end to end.

S start · J junction · G goal · T dead end · O other start

Stitching in DT: Yamagata et al., ICML 2023

Proposal: relabel returns with IQL

STEP 1

Learn with IQL

IQL fits Q(state, action) and V(state) on the logged data, with no unseen actions.

STEP 2

Relabel returns

New target = max(actual return, Q(state, action)). A candidate rule, to be tuned.

STEP 3

Train DT as usual

Same Transformer, same loss, better labels.

Why it should help. Q(state, action) estimates what the logged action can lead to, using good continuations from other trajectories, so a good first half is no longer penalized for a bad second half.

Experimental plan

What	Plan
Benchmarks	D4RL MuJoCo (HalfCheetah, Hopper, Walker2d; medium, medium-replay, medium-expert), plus Franka Kitchen
Methods	DT, QDT-style relabeling with Q-learning, IQL, and DT with IQL relabeling (mine)
Metric	Normalized D4RL score, mean and standard deviation over several seeds
Ablations	Context length, relabeling strength, dataset quality

> Reproduction first: my scores are checked against the published ones.

Benchmarks: Fu et al., D4RL, 2020

05 · WRAP-UP

Take-aways

1. Offline RL lets robots learn from logged data, with no risky trial and error.
2. Decision Transformer turns RL into sequence modeling: simple and stable, but it struggles to stitch.
3. My project reproduces it, then tests whether an IQL value function helps with stitching by relabeling returns.

Questions?